Lab 1 - GPS Driver and Data Analysis

EECE 5554 - Robotics: Navigation and Sensing

For this lab, a GPS device driver was written that reads $GPGGA sentences from the GNSS puck, converts latitude/longitude to UTM, and publishes it as a custom ROS2 message. Two datasets were then collected outside: one where the puck was kept still for about 11 minutes, and one where the puck was carried while walking in a straight-ish line for a few hundred meters. Below is the analysis of both datasets.

1. Stationary Data

The puck was kept in one spot for about 11 minutes, and 184 GPGGA fixes were recorded. Since the puck did not move, any spread in the readings is just GPS error/noise. Here are the basic stats:

Quantity	Value
Number of readings	184
Duration	~11.4 minutes (682 s)
Mean position (UTM)	(328053.14, 4687905.31) m
Std deviation (easting, northing)	(10.68 m, 6.49 m)
Mean distance from mean position	12.10 m
Max distance from mean position	16.06 m

So even while standing still, a single reading can be off by around 12-16 meters from the true spot. That is a pretty big error for something that did not move at all. The points below are also colored by time, to check whether the error was just random jitter around one spot, or if it was drifting somewhere over time.

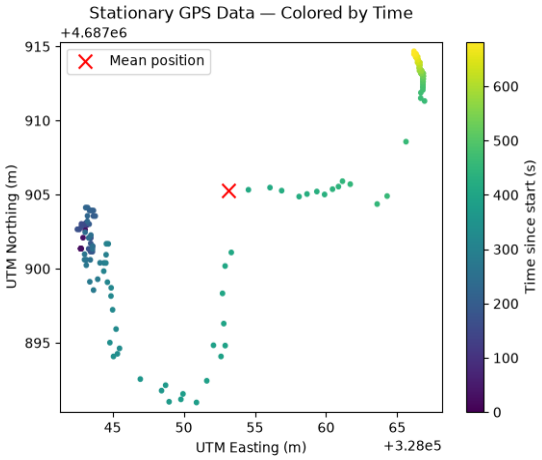


Figure 1: Stationary GPS readings, colored by time since the start of recording.

This plot shows the error was not just random noise. The points slowly moved from one side (dark, early points) to the other side (yellow, later points) over the 11 minutes. The puck was placed close to a building while recording this, so this is most likely multipath error - the GPS signal was bouncing off the building before reaching the puck, and this bounced signal changes as the satellites move across the sky, which makes the error drift instead of just randomly scatter.

Sources of error for stationary data:

- Multipath - signal reflecting off the nearby building (main reason here, based on the drift we saw)
- Satellite geometry changing over the 11 minutes (also called DOP)
- Normal receiver/atmospheric noise

2. Moving (Walking) Data

For the second part, the puck was carried while walking for a few minutes, mostly in one direction with a turn-around partway through. 232 fixes were recorded over about 231 seconds.

Quantity	Value
Number of readings	232
Duration	231 s (~3.85 min)
Straight-line displacement (start to end)	169.10 m
Total path length (sum of steps)	296.44 m
Mean speed	1.28 m/s
Max speed	2.05 m/s
Std deviation of speed	0.31 m/s

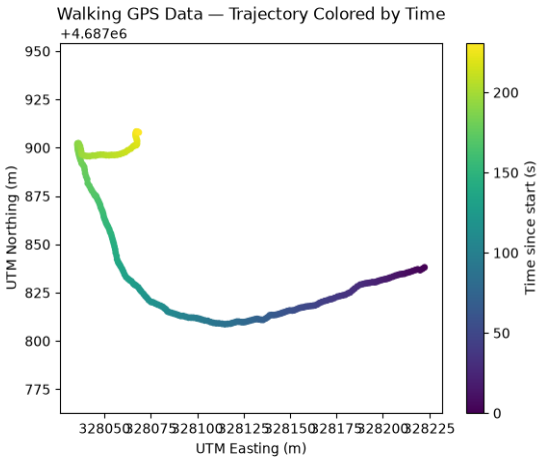


Figure 2: Walking path, colored by time since the start of the walk.

The path looks smooth and continuous, and the calculated mean speed (1.28 m/s) matches a normal walking speed, so the data looks trustworthy. The path length (296 m) is a bit more than the straight-line distance (169 m) because the walk was not perfectly straight the whole time, plus there is some GPS noise adding small zig-zags to the path.

Compared to the stationary case, the error while moving is harder to measure directly because there is no single 'true' point to compare against. But since the speed stayed in a reasonable, human-walking-pace range (0.5 to 2 m/s) without any sudden unrealistic jumps, the noise while moving does not seem worse than while standing still - if anything, GPS looks a bit more usable while moving, since the same slow drift seen in the stationary case does not show up here.

Sources of error for moving data:

- Small GPS position noise causing zig-zags in an otherwise straight path
- Turn-around point in the walk (shows up as a small loop in the plot)
- Same general GPS error sources as above (satellite geometry, atmosphere)

3. Conclusion

Stationary GPS error was around 12-16 meters and was not random - it drifted over time, most likely because I was near a building (multipath). Moving GPS data looked more reliable since the computed walking speed matched real walking speed closely. Overall this showed that GPS accuracy depends a lot on the surroundings (like nearby buildings) and not just on how long you wait for a fix.